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Confidence-aware refinement and learning for class imbalance in knowledge tracing

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ABSTRACT

Knowledge tracing models a learner's evolving mastery from sequential educational interactions and is a core component of adaptive learning systems. Despite substantial progress, diagnostic reliability remains constrained because heterogeneous learning behaviors are often compressed into a single latent state and imbalanced response distributions bias training toward the majority class. To address these challenges, we propose Confidence-Aware Refinement and Learning for Class Imbalance in Knowledge Tracing (CareKT), a hierarchical framework that combines global sequence modeling, multiple behavioral views, refinement guided by confidence, and optimization for class imbalance. CareKT preserves diverse local perspectives on a learner's trajectory, derives the primary estimate by averaging local probabilities across views, and uses these probabilities to refine local representations, while an auxiliary predictor integrating global and local evidence provides additional supervision. Focal loss further improves sensitivity to diagnostically important incorrect responses. Experiments on four benchmark datasets show that CareKT achieves the best area under the receiver operating characteristic curve (AUC) on three datasets and remains competitive on NIPS34. It also attains the best area under the precision–recall curve (AUCPRC) on ASSIST09, ASSIST12, and ASSIST15 among representative baselines. These findings indicate that jointly addressing representation entanglement and class imbalance improves diagnostic reliability. Our code is available at https://anonymous.4open.science/r/CareKT.

1. Introduction

Knowledge tracing (KT) is a fundamental task in intelligent tutoring systems and adaptive learning platforms. It models a learner's evolving knowledge state from historical interactions and predicts future performance on upcoming exercises. Such predictions support downstream educational decisions including resource recommendation, remediation, and timely intervention. Beyond average predictive accuracy, however, the practical value of a KT model also depends on whether it can provide diagnostically reliable signals about a learner's stable and vulnerable knowledge states, as well as the likelihood of future errors. Fig. 1 illustrates a typical KT scenario and the evolution of mastery across skills over time.

A recent survey provides a detailed overview of advances in KT [1]. Compared with early probabilistic formulations such as Bayesian Knowledge Tracing (BKT), recurrent architectures, memory-augmented models, and attention-based variants provide substantially stronger sequence modeling capacity [2–7]. Nevertheless, these advances do not fully resolve a central difficulty in learner modeling. Many existing approaches still encode complex learning behavior into a single latent state updated by relatively

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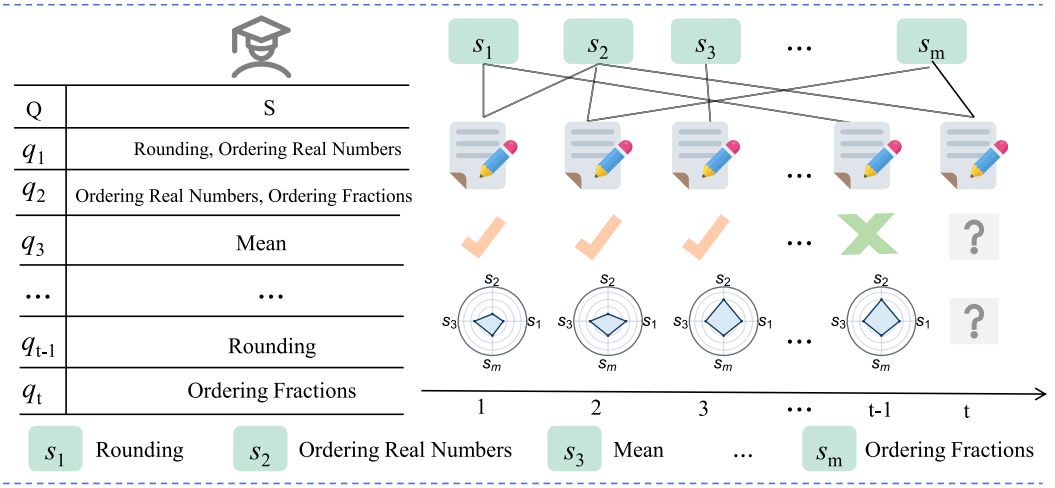


Fig. 1. Illustration of a knowledge tracing scenario. A student interacts with questions associated with different skills, and the learner state evolves over time as additional responses are observed. The radar chart provides a schematic view of changes in mastery for individual skills during this process.

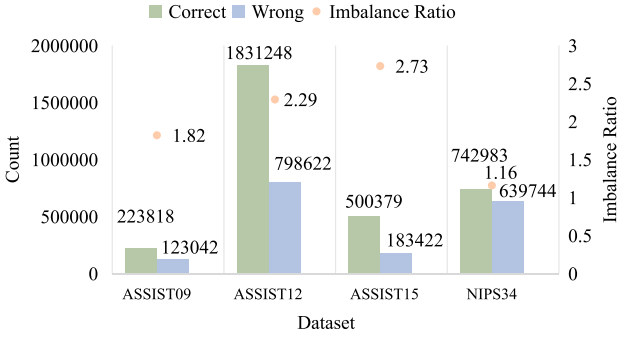


Fig. 2. Class imbalance ratios for the four benchmark KT datasets based on valid interaction records with responses labeled as correct or incorrect. The ratio is defined as $R = n_{\text{correct}}/n_{\text{incorrect}}$, where correct and incorrect responses correspond to the majority and minority classes, respectively. The resulting ratios are 1.82 for ASSIST09, 2.29 for ASSIST12, 2.73 for ASSIST15, and 1.16 for NIPS34.

homogeneous transition logic. Consequently, gains in predictive performance do not necessarily translate into comparable gains in diagnostic reliability.

A major source of this limitation is representation entanglement. An observed response is not determined by mastery alone. It may also reflect guessing, slipping, partial understanding, transient attention fluctuations, or noise in the interaction record. Recent KT studies increasingly model specific aspects of learner behavior more explicitly, including forgetting effects, question-centric cognition, response disentanglement, uncertainty, length stability, and robustness to noisy interactions [8–15]. When these heterogeneous factors are compressed into a single representation, the model may still struggle to distinguish stable knowledge from short-term irregularities. The resulting latent state becomes harder to interpret and more vulnerable to noisy evidence, especially in the absence of an explicit mechanism for calibrating local signals by their reliability. These limitations motivate our emphasis on diagnostic reliability through multi-view preservation and prediction-aware refinement of local behavioral evidence.

Class imbalance presents a second challenge in educational datasets [16]. Correct responses usually dominate incorrect responses, so standard training objectives are strongly influenced by the majority class. From a diagnostic perspective, this is problematic because learner errors often provide the most informative evidence for identifying weak concepts and planning intervention. A model that performs well on average but remains insensitive to minority error cases may therefore be less educationally valuable than its aggregate accuracy suggests. Accordingly, conventional metrics such as accuracy (ACC) are not sufficient on their own for diagnostic evaluation, especially under imbalanced response distributions [17]. The imbalance ratio is defined as the ratio between the majority and minority classes. In all four datasets, correct responses constitute the majority class and incorrect responses constitute the minority class, yielding $R = n_{\text{correct}}/n_{\text{incorrect}}$. Fig. 2 shows that all four benchmark datasets are imbalanced, with R ranging from 1.16 on NIPS34 to 2.73 on ASSIST15.

To address these issues, we propose Confidence-Aware Refinement and Learning for Class Imbalance in Knowledge Tracing (CareKT), a framework for knowledge tracing that combines confidence-aware refinement with learning for class imbalance.

CareKT follows a hierarchical design that progresses from global sequence modeling to local refinement. A sequential encoder first captures the overall learning trajectory. A multi-view behavioral decomposition module then maps this trajectory into several local perspectives so that diverse behavioral patterns can be modeled without being immediately collapsed into a single vector. Building on this decomposition, CareKT derives the primary estimate by averaging local probabilities across views and uses these probabilities to refine local representations, while an auxiliary global-local predictor provides additional supervision during training, thereby reducing the propagation of weakly supported noise. Finally, an objective for class imbalance strengthens the model's sensitivity to diagnostically important incorrect responses.

The contributions of this work are summarized as follows:

- CareKT is proposed as a hierarchical KT framework that jointly addresses representation entanglement and class imbalance within a unified formulation centered on diagnostic reliability.
- A confidence-aware refinement mechanism is introduced that derives the primary estimate by averaging local probabilities across views and uses these probabilities to refine local representations, while an auxiliary global-local predictor provides additional supervision during training.
- Across four benchmark datasets, CareKT achieves the highest area under the receiver operating characteristic curve (AUC) on three datasets and remains competitive on NIPS34. In terms of area under the precision–recall curve (AUCPRC), it outperforms the representative baselines on ASSIST09, ASSIST12, and ASSIST15 while remaining competitive on NIPS34.

2. Related work

2.1. Deep models for knowledge tracing

Relative to probabilistic formulations such as BKT, deep KT research mainly advances by using more expressive encoders for learner histories [1,2]. DKT and its extensions model learner trajectories as recurrent state transitions and incorporate regularization or forgetting effects [3,8,18]. DKVMN and SKVMN extend this paradigm with external memory for storing concept information, while SAKT, SAINT, and AKT use attention or Transformer architectures to retrieve relevant historical interactions over longer contexts [4–7,19]. These representative encoder families form the main architectural context for the experimental comparison in this work. Across these architectures, progress mainly comes from stronger history encoding and better modeling of long-range temporal dependencies.

Recent studies also examine the stability of these encoders under more demanding conditions. SparseKT improves robustness by restricting attention to a sparse subset of historical interactions, and stableKT studies how attention models generalize when training and test sequences differ substantially in length [13,20]. ReKT revisits KT from a model-design perspective and shows that strong performance can also be achieved with a lightweight architecture by jointly modeling knowledge states at the question, concept, and domain levels through its Forget-Response-Update framework [21]. These advances improve the strength and stability of the backbone while continuing to rely largely on a unified learner state, and they therefore provide limited support for explicitly preserving heterogeneous behavioral evidence.

2.2. Representation and reliability in knowledge tracing

KT representation learning has increasingly moved beyond the temporal order of interactions alone. Recent question-centric and contrastive approaches improve cognitive alignment and interpretability by constructing representations around question-level evidence and multiple expert views [9,15]. These studies show that KT benefits from richer evidence sources, but stronger contextual information does not by itself guarantee that the learned representations separate different causes of observed responses.

This limitation is addressed more directly in later work on representation quality. Beyond question-centric and contrastive representation learning, CRKT introduces response disentanglement with concept map information, and RobustKT explicitly separates cognitive patterns from error-prone data [10,11]. UKT and DTransformer further emphasize uncertainty and stability in learner representations, underscoring the need to distinguish persistent knowledge from transient interaction patterns shaped by uncertainty [12,14]. These studies highlight the heterogeneity of learner behavior while offering limited mechanisms for preserving multiple local views and calibrating them by predictive confidence before fusion.

2.3. Class imbalance and diagnostic evaluation

Response imbalance is common in educational data, and the broader literature on imbalanced learning has long shown that minority cases require dedicated treatment [16]. In KT, incorrect responses are usually less frequent than correct responses, but they often carry the most direct diagnostic evidence for remediation and intervention. Under this diagnostic objective, evaluation based only on accuracy or overall ranking is insufficient. AUCPRC, namely the area under the precision–recall curve, is particularly informative in this setting because it evaluates how well the model retrieves minority error cases across decision thresholds and is generally more sensitive to minority-class performance than receiver operating characteristic (ROC)-based summaries under imbalanced distributions [17].

At the optimization level, imbalance-aware strategies include re-sampling and re-weighting objectives, with focal loss emphasizing hard and minority examples by reducing the contribution of easy majority cases [22]. Within KT, however, class imbalance is

Table 1
Description of key symbols in the CareKT framework.

Symbol	Explanation
X	Interaction sequence $\{(q_1, r_1), \ldots, (q_T, r_T)\}$.
N	Total number of distinct concepts after preprocessing.
$\mathbf{x}_t$	Input embedding at step t: $\mathbf{x}_t = \mathbf{E}^{int}(q_t + r_t \times N)$.
$\mathbf{h}_t$	Global hidden state at step t produced by the gated recurrent unit (GRU) encoder.
$\mathbf{a}_k$	Trainable attention vector for view k.
$e_{t,k}$	Attention score of view k at step t: $e_{t,k} = \mathbf{a}_k^\top \mathbf{h}_t$.
m_t	Binary mask indicating whether step t is valid in the sequence.
s_t	Binary mask indicating whether prediction step t is valid for next-step supervision.
$\alpha_{t,k}$	Normalized attention weight assigned by view k at step t.
$\mathbf{v}_{t,k}$	View-specific representation: $\mathbf{v}_{t,k} = \alpha_{t,k}\mathbf{h}_t$.
$\mathbf{p}_{t,k}$	Concept-wise local prediction vector produced by view k at step t.
$p_{t,k}$	Local predictive probability for target concept q_{t+1} under view k.
$\hat{y}_t$	Primary predicted probability for target concept q_{t+1} via averaged multi-view prediction.
$\mathbf{g}_{t,k}$	Refinement signal projected from $\mathbf{p}_{t,k}$ and applied to view k.
$\mathbf{r}_{t,k}$	Refined feature: $\mathbf{r}_{t,k} = \mathbf{v}_{t,k} \odot \mathbf{g}_{t,k}$.
$\mathbf{R}_t$	Aggregated local feature via max pooling: $\mathbf{R}_t = \max_k(\mathbf{r}_{t,k})$.
$\mathbf{h}'_t$	Linearly transformed global hidden state.
$\tilde{\mathbf{p}}_t$	Auxiliary concept-wise prediction vector produced by the global-local predictor.
$\tilde{y}_t$	Auxiliary predicted probability for target concept q_{t+1}.
p_t^*	Probability aligned with the true label in focal loss.
$\mathcal{L}_{pred}$	Primary prediction loss implemented with binary cross-entropy (BCE) on averaged multi-view output $\hat{y}_t$.
$\mathcal{L}_{aux}$	Auxiliary refinement loss (BCE) on refined output $\tilde{y}_t$.
$\mathcal{L}_{focal}$	Focal loss for class imbalance: $-\frac{1}{T-1}\sum \alpha_f (1-p_t^*)^\gamma \log p_t^*$.
$\mathcal{L}$	Total objective: $\mathcal{L} = \mathcal{L}_{pred} + \lambda_{aux}\mathcal{L}_{aux} + \lambda_{foc}\mathcal{L}_{focal}$.

usually treated as secondary to the main architecture design and is often handled as an auxiliary training choice rather than as part of a diagnostic formulation that jointly considers representation learning, optimization, and evaluation. As a result, sequence modeling, representation reliability, and sensitivity to minority errors are usually studied separately. These gaps motivate a framework that jointly preserves multiple behavioral views, refines them according to confidence, and optimizes explicitly for diagnostically important minority errors.

3. Problem formulation

A learner's historical interaction sequence is denoted by

$$X = \{(q_1, r_1), (q_2, r_2), \ldots, (q_T, r_T)\}, \tag{1}$$

where q_t denotes the concept identifier at time step t after preprocessing, and $r_t \in \{0, 1\}$ denotes the corresponding response correctness. Here, $r_t = 1$ indicates a correct response and $r_t = 0$ indicates an incorrect response. This notation is kept sufficiently general to remain compatible with common preprocessing protocols used in public KT benchmarks.

The standard KT objective is to estimate the probability that the learner will answer the next interaction correctly given the observed history:

$$P(r_{t+1} = 1 \mid q_{t+1}, X_t), \tag{2}$$

where $X_t = \{(q_j, r_j)\}_{j=1}^{t}$ denotes the interaction history observed up to time step t. A KT model therefore needs to summarize past interactions into a latent learner representation that remains predictive for future performance.

This study considers KT from the perspective of *diagnostic reliability*, which is characterized by two complementary properties. The first is strong overall discrimination, namely the ability to separate likely correct responses from likely incorrect responses across the full distribution. The second is sensitivity to minority error cases, which is particularly important when the goal is to identify fragile knowledge states rather than merely fit the dominant response pattern. This definition motivates the joint use of AUC, ACC, and AUCPRC in the empirical evaluation.

For clarity, Table 1 summarizes the core symbols used throughout the remainder of the paper.

4. The CareKT model

CareKT is designed to improve diagnostic reliability by addressing representation entanglement and class imbalance. As shown in Fig. 3, the model adopts a hierarchical architecture that progresses from global sequence encoding to local refinement. The sequential encoder summarizes the interaction history into a global trajectory state. View-specific sequence weighting then decomposes this trajectory into multiple local perspectives. The predictive outputs of these views form an averaged primary estimate and provide refinement signals for an auxiliary global-local predictor. Training is guided by supervision on the primary output, auxiliary supervision on the refined predictor, and focal reweighting for class imbalance.

Algorithm 1 summarizes the training paradigm of CareKT.

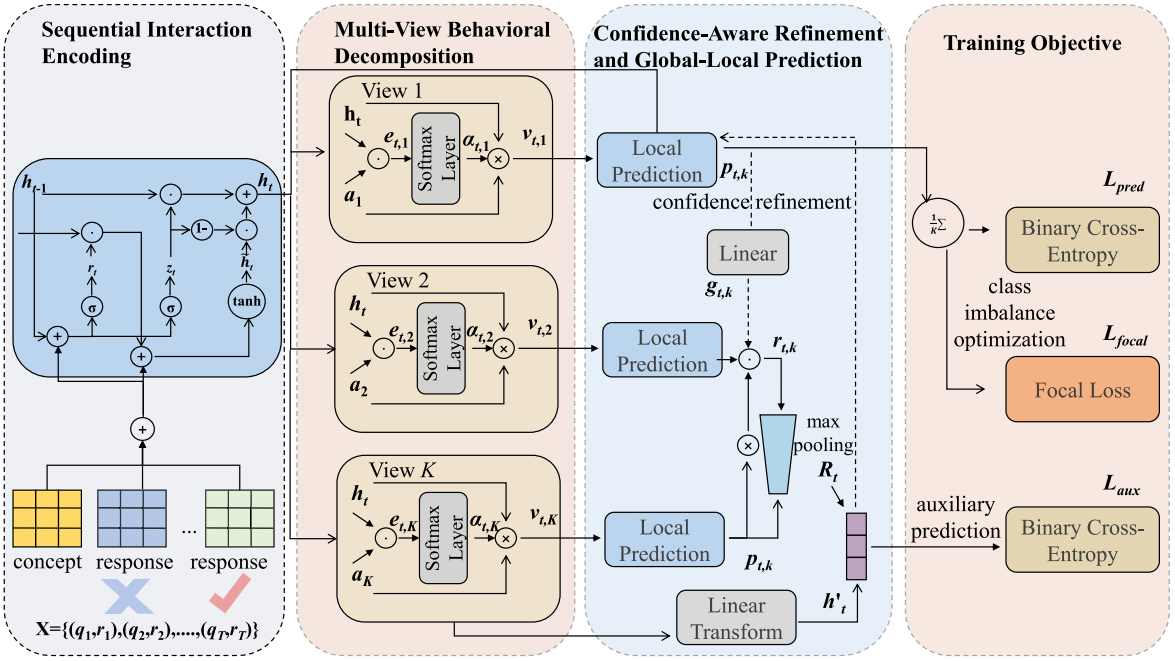


Fig. 3. Overview of CareKT. The model consists of sequential interaction encoding, multi-view behavioral decomposition, confidence-aware refinement, an auxiliary global-local predictor, and optimization with primary prediction loss, auxiliary refinement loss, and focal loss.

Algorithm 1 Training paradigm of CareKT.

Input: Interaction sequence $X = \{(q_t, r_t)\}_{t=1}^{T}$, next-step targets $\{q_{t+1}\}_{t=1}^{T-1}$, sequence masks $\{m_t\}_{t=1}^{T}$, valid prediction mask $\{s_t\}_{t=1}^{T-1}$, number of views K.
Output: Primary predictions $\hat{y}_{1:T-1}$ and training objective $\mathcal{L}$.

Encode interactions and obtain global hidden states $\mathbf{H} = \{\mathbf{h}_t\}_{t=1}^{T}$.
for each view $k = 1, \ldots, K$ **do**
  Compute causal masked attention weights $\alpha_{t,k}$ over the observed prefix up to step t using the sequence mask.
  for each prediction step $t = 1, \ldots, T-1$ with $s_t = 1$ **do**
    $\mathbf{v}_{t,k} = \alpha_{t,k}\mathbf{h}_t$.
    $\mathbf{p}_{t,k} = \sigma(\mathbf{W}_{p,k}\mathbf{v}_{t,k} + \mathbf{b}_{p,k})$, $p_{t,k} = [\mathbf{p}_{t,k}]_{q_{t+1}}$.
    $\mathbf{g}_{t,k} = \mathbf{W}_{r,k}\mathbf{p}_{t,k} + \mathbf{b}_{r,k}$.
    $\mathbf{r}_{t,k} = \mathbf{v}_{t,k} \odot \mathbf{g}_{t,k}$.
  end for
end for
for each prediction step $t = 1, \ldots, T-1$ with $s_t = 1$ **do**
  $\hat{y}_t = \frac{1}{K}\sum_{k=1}^{K} p_{t,k}$.
  $\mathbf{R}_t = \max_k(\mathbf{r}_{t,k})$.
  $\mathbf{h}'_t = \mathbf{W}_h\mathbf{h}_t + \mathbf{b}_h$.
  $\tilde{\mathbf{p}}_t = \sigma\big(\mathbf{W}_{fus}(\mathbf{h}'_t \odot \mathbf{R}_t) + \mathbf{b}_{fus}\big)$, $\tilde{y}_t = [\tilde{\mathbf{p}}_t]_{q_{t+1}}$.
end for
Obtain $\mathcal{L}_{pred}$, $\mathcal{L}_{aux}$, and $\mathcal{L}_{focal}$ over valid prediction steps.
$\mathcal{L} = \mathcal{L}_{pred} + \lambda_{aux}\mathcal{L}_{aux} + \lambda_{foc}\mathcal{L}_{focal}$.
Update model parameters by minimizing $\mathcal{L}$ with Adam.
return $\hat{y}_{1:T-1}, \mathcal{L}$.

4.1. Sequential interaction encoding

The sequential interaction encoding module provides the backbone representation of a learner's trajectory. Let N denote the number of concepts after preprocessing. CareKT uses an interaction embedding matrix $\mathbf{E}^{int} \in \mathbb{R}^{2N \times d}$ to jointly encode the practiced

concept and the observed response. Each historical interaction (q_t, r_t) is mapped to

$$\mathbf{x}_t = \mathbf{E}^{int}(q_t + r_t \times N), \tag{3}$$

where $\mathbf{x}_t \in \mathbb{R}^d$ is the interaction embedding at time step t. This formulation assigns distinct embeddings to correct and incorrect interactions for the same concept.

The sequence of interaction embeddings is then processed by a gated recurrent unit (GRU), which updates the global hidden state according to

$$\mathbf{h}_t = \mathrm{GRU}(\mathbf{x}_t, \mathbf{h}_{t-1}). \tag{4}$$

The resulting hidden sequence $\mathbf{H} = \{\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_T\}$ summarizes the macroscopic evolution of the learner's trajectory and serves as the input to the subsequent view-specific prediction modules.

4.2. Multi-view behavioral decomposition

Although the global hidden state is effective for summarizing temporal context, a single vector may still entangle heterogeneous behavioral factors such as stable mastery, transient uncertainty, guessing, and slipping. CareKT therefore introduces K parallel local views, each intended to capture a distinct aspect of the learner's historical behavior. Here, K denotes the number of parallel local views and is treated as a tunable hyperparameter rather than an adaptive latent variable. Each view is parameterized by an independent attention vector that assigns salience weights over the encoded sequence.

For each time step t and each view $k \in \{1, \ldots, K\}$, the attention score is computed as

$$e_{t,k} = \mathbf{a}_k^\top \mathbf{h}_t, \tag{5}$$

where $\mathbf{a}_k$ is the trainable attention vector for view k. Let $m_t \in \{0, 1\}$ denote the sequence mask at time step t. The corresponding normalized attention weight is

$$\alpha_{t,k} = \frac{\exp(e_{t,k})\, m_t}{\sum_{\tau=1}^{t} \exp(e_{\tau,k})\, m_\tau}. \tag{6}$$

The view-specific representation at time step t is then defined by reweighting the hidden state:

$$\mathbf{v}_{t,k} = \alpha_{t,k} \mathbf{h}_t. \tag{7}$$

Each $\mathbf{v}_{t,k}$ therefore represents the contribution of time step t under view k, and the collection of all views forms a richer decomposition of the learner state than a single global sequence alone. This design realizes multi-view decomposition through view-specific sequence weighting, with each view assigning its own salience pattern over the encoded trajectory.

Because different views use independent attention parameters, they can learn different weighting profiles over the same encoded trajectory. These views are latent rather than manually tied to predefined categories such as guessing, slipping, or mastery. The shared recurrent backbone preserves temporal continuity, while the view-specific weighting mechanism provides complementary local emphases over the interaction history instead of collapsing all evidence into a single channel.

4.3. Confidence-aware refinement

The refinement stage uses local predictive outputs to modulate each view before aggregation. The modulation signal is given by the view-specific predictive probabilities. Dropout is omitted here for notational simplicity. For each view representation, the model first computes a concept-wise local prediction vector

$$\mathbf{p}_{t,k} = \sigma(\mathbf{W}_{p,k} \mathbf{v}_{t,k} + \mathbf{b}_{p,k}), \tag{8}$$

where $\mathbf{p}_{t,k} \in [0, 1]^N$. The scalar probability assigned by view k to the next target concept is obtained by selecting the component indexed by q_{t+1}:

$$p_{t,k} = [\mathbf{p}_{t,k}]_{q_{t+1}}. \tag{9}$$

The averaged multi-view prediction,

$$\hat{y}_t = \frac{1}{K} \sum_{k=1}^{K} p_{t,k}, \tag{10}$$

serves as the primary output used at inference time. To refine local features, each concept-wise prediction vector is projected back to the hidden space:

$$\mathbf{g}_{t,k} = \mathbf{W}_{r,k} \mathbf{p}_{t,k} + \mathbf{b}_{r,k}, \tag{11}$$

where $\mathbf{g}_{t,k} \in \mathbb{R}^d$. The refined local representation is then

$$\mathbf{r}_{t,k} = \mathbf{v}_{t,k} \odot \mathbf{g}_{t,k}, \tag{12}$$

where $\odot$ denotes element-wise multiplication. This operation allows each view to modulate its local representation according to its own predictive profile over concepts. The local prediction vector summarizes the concept-level outcomes favored by the current view, and its projection back to the hidden space yields a feature-wise refinement signal aligned with that profile.

4.4. Global-local prediction

After refinement, the model aggregates local evidence across views and combines it with the global trajectory representation. First, CareKT uses max pooling to retain the most salient local signal along each feature dimension:

$$\mathbf{R}_t = \max_{k\in\{1,\dots,K\}} \mathbf{r}_{t,k}. \tag{13}$$

This asymmetric pooling choice preserves the strongest refined local evidence instead of averaging away salient signals across views.

The global hidden state is then linearly transformed:

$$\mathbf{h}'_t = \mathbf{W}_h \mathbf{h}_t + \mathbf{b}_h, \tag{14}$$

and the refined global-local interaction is mapped to an auxiliary concept-wise prediction vector:

$$\tilde{\mathbf{p}}_t = \sigma\left(\mathbf{W}_{fus}(\mathbf{h}'_t \odot \mathbf{R}_t) + \mathbf{b}_{fus}\right), \tag{15}$$

where $\tilde{\mathbf{p}}_t \in [0,1]^N$. The corresponding auxiliary prediction for the target concept is

$$\tilde{y}_t = [\tilde{\mathbf{p}}_t]_{q_{t+1}}. \tag{16}$$

During training, this global-local predictor is used only as an auxiliary refined predictor, whereas the averaged multi-view prediction $\hat{y}_t$ remains the primary output at inference time. This separation keeps $\hat{y}_t$ as the inference-time output while using the auxiliary branch to regularize the interaction between global and local evidence during training.

4.5. Training objective

CareKT is trained with a three-term objective. The primary loss directly supervises the inference-time predictor, the auxiliary loss regularizes the refined global-local predictor during training, and the focal term increases sensitivity to hard and minority error cases. Because the model predicts r_{t+1} from the history observed up to step t, all supervision terms are indexed from $t = 1$ to $T - 1$. The first term is the standard binary cross-entropy loss applied to the averaged multi-view prediction:

$$\mathcal{L}_{pred} = -\frac{1}{T-1}\sum_{t=1}^{T-1}\left[r_{t+1}\log\hat{y}_t + (1-r_{t+1})\log(1-\hat{y}_t)\right]. \tag{17}$$

This term directly supervises the primary output used at inference time.

The second term applies the same binary supervision to the auxiliary refined prediction:

$$\mathcal{L}_{aux} = -\frac{1}{T-1}\sum_{t=1}^{T-1}\left[r_{t+1}\log\tilde{y}_t + (1-r_{t+1})\log(1-\tilde{y}_t)\right]. \tag{18}$$

This auxiliary term encourages the refined global-local predictor to remain predictive without altering the role of the primary output used at inference time.

To improve sensitivity to hard and minority examples, especially diagnostically important learner errors, CareKT further introduces a term based on focal loss [22]. In the present formulation, this focal term is also applied to the primary averaged prediction:

$$p_t^* = r_{t+1}\hat{y}_t + (1-r_{t+1})(1-\hat{y}_t). \tag{19}$$

The focal loss for class imbalance is defined as

$$\mathcal{L}_{focal} = -\frac{1}{T-1}\sum_{t=1}^{T-1}\alpha_f(1-p_t^*)^{\gamma}\log(p_t^*), \tag{20}$$

where α_f is a global scaling coefficient for the focal term and γ controls the degree of down-weighting for well-classified examples, thereby emphasizing harder cases. The total training objective is

$$\mathcal{L} = \mathcal{L}_{pred} + \lambda_{aux}\mathcal{L}_{aux} + \lambda_{foc}\mathcal{L}_{focal}, \tag{21}$$

where λ_{aux} and λ_{foc} control the relative contributions of the auxiliary refinement term and the focal term, respectively. In the training objective, these coefficients correspond to the weights of the auxiliary BCE branch and the focal loss branch. Architecturally, CareKT addresses representation entanglement and unreliable local evidence, while the objective emphasizes diagnostically important minority error cases.

5. Experimental setup

CareKT is evaluated on four public benchmark datasets and compared with a diverse set of KT baselines. The experiments examine overall predictive performance, the contribution of individual components, and robustness under class imbalance.

Table 2
Statistics of the four benchmark knowledge tracing datasets. The row for the imbalance ratio reports the values summarized in Fig. 2, where $R = n_{\text{correct}}/n_{\text{incorrect}}$. The row for sequence length reports the truncation lengths used in the finalized main experimental protocol. A dash indicates that the corresponding statistic is unavailable for a given dataset release and preprocessing setting.

Items	ASSIST09	ASSIST12	ASSIST15	NIPS34
# User	4661	33,568	19,292	9401
# Question	17,737	53,070	–	948
# Concept	123	265	100	57
# Interaction	337,415	2,709,568	682,789	1,399,470
Seq Len	88	99	36	200
Avg c per q	1.197	1	–	1
Max concept	4	1	1	2
Imbalance ratio R	1.82	2.29	2.73	1.16

5.1. Datasets and preprocessing

Experiments are conducted on four widely used KT benchmarks, namely ASSIST09, ASSIST12, and ASSIST15 from ASSISTments,[1] together with NIPS34 from the NeurIPS 2020 Education Challenge.[2] These datasets differ markedly in scale, sparsity, sequence length, and response imbalance, and they cover both tutoring and diagnostic settings. Table 2 summarizes the benchmark characteristics and the sequence settings adopted in the experiments.

- **ASSIST09:** Collected from the ASSISTments platform during the 2009–2010 school year. The updated release is adopted because it corrects issues in the original version such as duplicated records.
- **ASSIST12:** Derived from the 2012–2013 Skill Builder problem sets of the same platform. It is substantially larger and sparser than ASSIST09, which makes it useful for testing model behavior under heavier data volume and longer logs.
- **ASSIST15:** A later ASSISTments benchmark characterized by comparatively short sequences and limited metadata in commonly used releases, making it a useful complement to the earlier ASSIST datasets.
- **NIPS34:** Released for the NeurIPS 2020 Education Challenge and derived from the Eedi platform. Unlike the ASSIST benchmarks, it consists of diagnostic multiple-choice mathematics questions and therefore provides a complementary evaluation setting.

Following the standard protocol adopted in PyKT [23], we remove records with missing attributes, discard students with very short histories, split multi-skill questions when required by the benchmark format, and pad or truncate sequences for mini-batch training. For the class-distribution analysis, we retain valid binary-response interaction records and define the imbalance ratio as $R = n_{\text{correct}}/n_{\text{incorrect}}$, where correct and incorrect responses correspond to the majority and minority classes, respectively. Fig. 2 summarizes the resulting class distributions, and the corresponding ratios are also listed in Table 2. After preprocessing, interactions are represented at the concept level throughout the proposed formulation, so q_t denotes a concept identifier in all experiments. Under the finalized main experimental protocol, the truncation lengths are 88, 99, 36, and 200 for ASSIST09, ASSIST12, ASSIST15, and NIPS34, respectively. Model evaluation follows the PyKT protocol with five training and validation splits. In each run, one fold from the preprocessed training and validation split is used for validation and the remaining four folds are used for training under the same PyKT preprocessing pipeline. The validation fold is used only for early stopping and hyperparameter selection, after which the selected checkpoint is evaluated on the benchmark test split. Final results are reported as averages over the five runs obtained by rotating the validation fold.

5.2. Baselines

CareKT is compared with a diverse set of representative KT baselines spanning historically influential models, recent competitive models, and methods with additional inductive biases such as difficulty modeling or temporal dynamics. To avoid restricting the evaluation to a single architectural trend, these baselines are grouped into four model families.

5.2.1. RNN-based models

- **DKT** [3]: The canonical recurrent KT model for sequential learner modeling.
- **DKT+** [18]: A regularized extension of DKT designed to improve predictive stability.
- **DKT-F** [8]: A forgetting-aware recurrent variant that incorporates temporal decay effects.
- **Deep-IRT** [24]: A recurrent model that combines KT with item response theory for greater interpretive structure.
- **AT-DKT** [25]: A recurrent KT model enhanced with auxiliary tasks for question tagging and individualized prior knowledge prediction.

[1] https://sites.google.com/site/assistmentsdata.
[2] https://eedi.com/projects/neurips-education-challenge.

Table 3
Final dataset-specific hyperparameter settings for CareKT. Here, α_f is set to the class imbalance ratio $R = n_{\text{correct}}/n_{\text{incorrect}}$ for each dataset, and γ is fixed to 2.0 across datasets.

Dataset	K	λ_{aux}	λ_{foc}	α_f	γ
ASSIST09	4	0.9	5.0	1.82	2.0
ASSIST12	8	2.0	3.0	2.29	2.0
ASSIST15	8	2.0	2.5	2.73	2.0
NIPS34	6	2.5	4.0	1.16	2.0

5.2.2. Memory-based models

- **DKVMN** [4]: A key–value memory network that explicitly tracks concept-level knowledge states.
- **SKVMN** [19]: A memory-augmented model designed to capture longer-range dependencies in knowledge evolution.

5.2.3. Attention-based models

- **SAKT** [5]: A self-attention model that emphasizes relevant historical interactions.
- **SAINT** [6]: A Transformer-style KT architecture with separate treatment of exercises and responses.
- **AKT** [7]: A context-aware attentive model with monotonic temporal dynamics.
- **ATKT** [26]: An attention-based model enhanced with adversarial training.
- **SimpleKT** [27]: A simplified attention baseline with competitive empirical performance.
- **SparseKT** [20]: A sparsity-aware attention model that focuses on influential historical interactions.

5.2.4. Specialized models

- **KQN** [28]: A model that represents student learning activities as knowledge state vectors and models their interactions with skill vectors for prediction.
- **HawkesKT** [29]: A temporal point-process model that explicitly captures cross-interaction temporal effects.
- **DIMKT** [30]: A model that incorporates question difficulty information into KT.
- **RobustKT** [11]: A robustness-oriented KT model that explicitly decouples cognitive patterns from error-prone interaction data to reduce the influence of noisy responses.
- **ReKT** [21]: A lightweight KT model that traces student knowledge at the question, concept, and domain levels under a Forget-Response-Update framework.

5.3. Implementation details

All models are implemented in PyKT [23]. For fair comparison, baseline experiments are reproduced with the corresponding PyKT entry scripts under the same preprocessing pipeline and data folds as CareKT. Model selection is performed according to validation AUC, and the executable scripts used to reproduce the reported results are included in the project repository. Models are optimized with Adam for up to 200 epochs, with early stopping if validation AUC does not improve for 10 consecutive epochs. All experiments are conducted on a computing cluster equipped with NVIDIA RTX 3090 GPUs. AUC, ACC, and AUCPRC are reported, with AUC treated as the primary metric. For CareKT, the tuned hyperparameters include the number of views K and the loss coefficients λ_{aux} and λ_{foc}, which are selected on the validation set for each dataset and then fixed for all runs on that dataset. Here, λ_{aux} and λ_{foc} denote the coefficients of the auxiliary refinement branch and the focal loss branch, whereas α_f and γ denote the focal loss scaling and focusing parameters. For each dataset, α_f is set to the imbalance ratio $R = n_{\text{correct}}/n_{\text{incorrect}}$, and γ is fixed to 2.0 across datasets rather than tuned separately. Table 3 summarizes the resulting dataset-specific hyperparameter settings. To assess robustness beyond single-seed reporting, a multi-seed significance analysis is further conducted using the shared seed set $\{42, 52, 62, 72, 82\}$.

6. Results and analysis

The proposed CareKT framework is evaluated across diverse educational data distributions to address four core research questions:

- **RQ1 (Overall Performance):** How does CareKT compare with competitive baselines in predictive discrimination across diverse data regimes?
- **RQ2 (Diagnostic Robustness):** How does the focal optimization strategy for class imbalance enhance the identification of critical minority error cases?
- **RQ3 (Component Ablation):** What are the isolated and synergistic contributions of the multi-view decomposition, confidence-aware refinement, and focal optimization components?
- **RQ4 (Interpretability & Sensitivity):** How does the architecture with multiple views realize behavioral decomposition and how sensitive is it to key design choices?

In addition to these four research questions, we also include a focused objective-stability analysis to examine whether the composite loss exhibits unstable interactions during optimization.

Table 4
Performance comparison on four benchmark datasets. For literature baselines, reference numbers and publication years are reported next to the method names. Higher values are better for all metrics, as indicated by ↑. Best results are shown in **bold**.

Methods	Year	ASSIST09		ASSIST12		ASSIST15		NIPS34	
		AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑
DKT [3]	2015	0.7497	0.7205	0.7306	0.7340	0.7249	0.7505	0.7666	0.7017
DKT+ [18]	2018	0.7512	0.7225	0.7312	0.7337	0.7274	0.7512	0.7661	0.7013
DKT-F [8]	2019	–	–	0.7320	0.7351	–	–	0.7651	0.7008
KQN [28]	2019	0.6993	0.6579	0.7280	0.7330	0.7246	0.7501	0.7632	0.6985
DKVMN [4]	2017	0.7424	0.7177	0.7236	0.7315	0.7213	0.7501	0.7545	0.6906
Deep-IRT [24]	2019	0.7428	0.7182	0.7265	0.7327	0.7211	0.7211	0.7627	0.6973
SKVMN [19]	2019	0.7205	0.7112	0.6907	0.7188	0.7033	0.7443	0.7247	0.6662
SAKT [5]	2019	0.7222	0.7033	0.7061	0.7233	0.6975	0.7450	0.7466	0.6836
ATKT [26]	2021	0.7371	0.7150	0.7254	0.7311	0.7200	0.7467	0.7559	0.6917
SAINT [6]	2020	0.6933	0.6937	0.6698	0.7135	0.6520	0.7343	0.7802	0.7124
AKT [7]	2020	0.7841	0.7414	0.7802	0.7582	0.7261	0.7517	0.7995	0.7286
HawkesKT [29]	2021	0.7300	0.7066	0.7483	0.7403	–	–	0.6003	0.6141
DIMKT [30]	2022	0.7633	0.7292	0.7698	0.7512	–	–	0.8002	0.7295
AT-DKT [25]	2023	0.7501	0.7209	0.7351	0.7363	–	–	0.7770	0.7115
SimpleKT [27]	2023	0.7736	0.7270	0.7734	0.7520	0.7234	0.7504	0.7993	0.7293
SparseKT [20]	2023	0.7688	0.7263	0.7712	0.7545	0.7224	0.7505	0.7965	0.7260
RobustKT [11]	2025	0.7841	0.7402	0.7793	0.7567	–	–	0.8001	0.7299
ReKT [21]	2024	0.7813	0.7349	0.7774	0.7562	–	–	**0.8015**	**0.7304**
CareKT	This work	**0.8137**	**0.7641**	**0.8396**	**0.7791**	**0.8343**	**0.8027**	0.7997	0.7262

Note: The symbol '–' indicates that the corresponding result is unavailable under the experimental setting considered here or that the model is inapplicable because required input features are unavailable in the corresponding dataset.

6.1. Overall performance (RQ1)

Table 4 summarizes the predictive discrimination of CareKT against the evaluated baselines. CareKT achieves the highest AUC on three of the four benchmark datasets. Relative to the best-performing baseline on each dataset, the AUC margin is 0.0296 on ASSIST09, 0.0594 on ASSIST12, and 0.1069 on ASSIST15, while on NIPS34 CareKT is 0.0018 below the best-performing baseline. The improvement is particularly pronounced on ASSIST12 and ASSIST15, while the NIPS34 result remains competitive, indicating that the model remains effective across both tutoring-style logs and longer diagnostic sequences.

The gains are not confined to a single model family. By outperforming attention-based baselines such as SAINT, AKT, and SimpleKT on ASSIST09, ASSIST12, and ASSIST15, CareKT shows that explicit multi-view decomposition and refinement guided by predictive probabilities can improve learner representations beyond sequence encoding alone. CareKT also exceeds the recent robustness-oriented baseline RobustKT and the lightweight baseline ReKT on ASSIST09 and ASSIST12, while the NIPS34 comparison indicates a competitive but not dominant outcome against the best-performing task-specific baselines. These comparisons indicate that the gains are not explained solely by decoupling noisy responses or by simplifying the tracing architecture. The NIPS34 comparison with DIMKT and ReKT further suggests that the proposed refinement mechanism remains competitive even when question difficulty or lightweight multi-level inductive biases are effective.

Although AUC is the primary metric, CareKT also achieves the highest ACC on ASSIST09, ASSIST12, and ASSIST15 under the same protocol, while remaining slightly below the best-performing NIPS34 baselines. The AUC and ACC results are therefore broadly consistent: CareKT obtains the best results on the three ASSIST benchmarks and remains competitive, though not dominant, on NIPS34. Preserving decomposed behavioral evidence improves both ranking quality and thresholded prediction accuracy.

6.2. Representative AUCPRC comparison

To further examine diagnostic robustness, we compare CareKT with five representative baselines in terms of AUCPRC, namely DKT, AKT, DIMKT, ReKT, and RobustKT. The reported results are averages over the five runs obtained by rotating the validation fold on all four benchmark datasets. Fig. 4 summarizes the AUCPRC results on the four benchmark datasets. CareKT achieves the highest AUCPRC on ASSIST09, ASSIST12, and ASSIST15, while remaining competitive on NIPS34. Relative to the best-performing representative baseline on each dataset, the margin is 0.0191 on ASSIST09, 0.0377 on ASSIST12, and 0.0557 on ASSIST15, whereas on NIPS34 CareKT is 0.0080 below ReKT. The improvement is most pronounced on ASSIST12 and ASSIST15, while the NIPS34 result indicates competitive identification of minority error cases, although it is not the best among the representative baselines considered here. These AUCPRC results indicate that the proposed framework improves the ranking of diagnostically important minority error cases on most benchmarks. DIMKT, ReKT, and RobustKT are not applicable to ASSIST15 under the concept-only release used in this study, and are therefore marked as 'N/A' in the figure.

To complement the AUCPRC comparison, Table 5 reports precision, recall, and F1-score for identifying incorrect responses, where an instance is predicted as incorrect when its predicted correctness probability is below 0.5. Incorrect responses are treated as the positive class. CareKT obtains the highest precision on ASSIST09, ASSIST12, and ASSIST15 and the highest F1-score on ASSIST09 and ASSIST15. On NIPS34, CareKT remains competitive, while ReKT records the highest F1-score among the representative baselines.



Fig. 4. Comparison with representative baselines in terms of AUCPRC on four benchmark datasets. Results are averaged over the five runs obtained by rotating the validation fold. Higher values indicate better identification of samples from the minority class, namely incorrect responses. 'N/A' denotes model-dataset pairs that are inapplicable under the corresponding dataset release and input setting.

Table 5
Precision, recall, and F1-score for identifying incorrect responses. Incorrect responses are treated as the positive class, and predictions with correctness probability below 0.5 are classified as incorrect. Results are averaged over the five runs obtained by rotating the validation fold. Higher values are better for all metrics. Metrics are computed for each run before averaging, so the reported F1-score is not derived from the rounded mean precision and recall shown in the table. Best values within each dataset are shown in **bold**. Ties after rounding are also bolded.

Dataset	Model	Precision↑	Recall↑	F1-score↑
ASSIST09	DKT	0.6705	0.4485	0.5372
	AKT	0.6752	0.4719	0.5553
	DIMKT	0.6616	0.5010	0.5700
	ReKT	0.6815	0.5093	0.5781
	RobustKT	0.6809	**0.5259**	0.5931
	CareKT	**0.7535**	0.5137	**0.6106**
ASSIST12	DKT	0.6356	0.2883	0.3966
	AKT	0.6478	0.4088	0.5013
	DIMKT	0.6387	0.4109	0.4998
	ReKT	0.6708	0.3899	0.4928
	RobustKT	0.6411	**0.4539**	**0.5314**
	CareKT	**0.7922**	0.3703	0.5039
ASSIST15	DKT	0.5754	0.2064	0.3037
	AKT	0.5712	0.2105	0.3076
	DIMKT	–	–	–
	ReKT	–	–	–
	RobustKT	–	–	–
	CareKT	**0.8667**	**0.5348**	**0.6613**
NIPS34	DKT	0.6597	0.6918	0.6753
	AKT	0.6749	**0.7391**	0.7055
	DIMKT	0.6873	0.7273	0.7066
	ReKT	**0.6883**	0.7344	**0.7104**
	RobustKT	**0.6883**	0.7318	0.7093
	CareKT	**0.6883**	0.7168	0.7022

Note: The symbol '–' indicates that the corresponding model is inapplicable under the ASSIST15 concept-only input setting used in this study.

6.3. Multi-seed significance analysis

To further assess statistical robustness, we conduct a multi-seed significance analysis under the dataset-specific hyperparameter settings described above using the shared seed set $\{42, 52, 62, 72, 82\}$. The role of Table 6 is different from that of Table 4. Specifically, Table 4 reports the main benchmark results averaged over the five runs obtained by rotating the validation fold, whereas Table 6 reports the mean ± standard deviation of AUC over multiple random seeds on a fixed data split, namely Fold 0, together with paired t-test results against multiple competitive baselines whenever applicable. Because Table 4 averages over rotated validation folds while Table 6 averages across seeds on a single fixed fold, the absolute AUC values in the two tables are not expected to match exactly. The completed comparisons show a clear advantage of CareKT over AKT, RobustKT, and ReKT on ASSIST09 and ASSIST12,

Table 6
Multi-seed significance comparison on AUC. Mean ± standard deviation across five random seeds on the fixed Fold 0 split are reported. Higher AUC and positive ΔAUC indicate better CareKT performance relative to the comparison baseline. ΔAUC denotes the difference between CareKT and the comparison baseline, and '–' indicates that the model is inapplicable under the current setting.

Dataset	Comparison baseline	Baseline AUC↑	CareKT AUC↑	ΔAUC↑	p-value
ASSIST09	AKT	0.7638 ± 0.0048	**0.8114 ± 0.0038**	+0.0476	<0.001
	RobustKT	0.7837 ± 0.0006		+0.0277	<0.001
	ReKT	0.7821 ± 0.0004		+0.0293	<0.001
ASSIST12	AKT	0.7691 ± 0.0018	**0.8543 ± 0.0127**	+0.0852	<0.001
	RobustKT	0.7766 ± 0.0004		+0.0777	<0.001
	ReKT	0.7743 ± 0.0004		+0.0800	<0.001
ASSIST15	AKT	0.7230 ± 0.0005	**0.8324 ± 0.0070**	+0.1094	<0.001
	RobustKT	–		–	–
	ReKT	–		–	–
NIPS34	DIMKT	0.7983 ± 0.0005	0.8013 ± 0.0089	+0.0030	0.5042
	RobustKT	0.8001 ± 0.0006		+0.0012	0.7832
	ReKT	**0.8016 ± 0.0010**		−0.0003	0.9451

Note: ASSIST15 does not support ReKT and RobustKT under the current input setting, so those entries are marked as '–'.

Table 7
Computational cost comparison between CareKT and the comparison baseline on each dataset. '#Params' denotes the number of trainable parameters, 'Train Time' denotes average seconds per epoch, and 'Inference' denotes average milliseconds per batch. Lower values are better for all three cost metrics. All measurements are obtained under the same settings as the main experiments on a single NVIDIA RTX 3090 GPU with batch size 64.

Dataset	Model	#Params↓	Train time↓ (s/epoch)	Inference↓ (ms/batch)
ASSIST09	AKT	1,528,931	4.51	31.39
ASSIST09	CareKT	**1,204,327**	**3.43**	**14.36**
ASSIST12	AKT	1,709,672	34.53	31.38
ASSIST12	CareKT	**863,569**	**28.19**	**18.16**
ASSIST15	AKT	1,410,329	17.78	26.30
ASSIST15	CareKT	**1,346,692**	**17.45**	**18.23**
NIPS34	DIMKT	**400,128**	71.13	220.98
NIPS34	CareKT	1,077,647	**6.60**	**18.34**

and over AKT on ASSIST15, with all of these paired comparisons satisfying $p < 0.001$. On NIPS34, the differences between CareKT and DIMKT ($p = 0.5042$), RobustKT ($p = 0.7832$), and ReKT ($p = 0.9451$) are all not significant, which is consistent with the relatively small gap between the best-performing models in Table 4. Overall, the multi-seed analysis supports a clear advantage of CareKT on the three ASSIST benchmarks and a competitive outcome on NIPS34.

6.4. Computational cost comparison

CareKT is further compared with a representative competitive baseline on each dataset in terms of trainable parameter count, per-epoch training time, and inference latency. In this efficiency analysis, the same dataset-specific settings as in the main experiments are used for CareKT, together with AKT on ASSIST09, ASSIST12, and ASSIST15, and DIMKT on NIPS34. All measurements are conducted on the same NVIDIA RTX 3090 GPU with batch size 64. Training time is averaged over two timed epochs after one warm-up epoch, and inference latency is averaged over up to 20 timed test batches after five warm-up batches.

Table 7 shows that CareKT is more parameter-efficient and faster than AKT on ASSIST09, ASSIST12, and ASSIST15. On NIPS34, CareKT has more trainable parameters than DIMKT, but its training and inference remain markedly faster under the same measurement protocol. This comparison further indicates that runtime efficiency is not determined by parameter count alone. Although DIMKT has fewer trainable parameters on NIPS34, its measured training and inference times are higher than those of CareKT under the same implementation and evaluation protocol. The additional multi-view and refinement components therefore do not impose a substantial runtime penalty in the present implementation.

6.5. Objective stability analysis

To examine whether the composite objective introduces unstable interactions among loss terms, we further analyze the training dynamics of CareKT under the finalized dataset-specific settings. Fig. 5 plots the total training loss together with its three components, namely the primary prediction loss L_{pred}, the auxiliary refinement loss L_{aux}, and the focal loss term L_{focal}, as well as the validation AUC and AUCPRC trajectories on all four datasets. Across datasets, the total loss and all component losses generally

Table 8
Pairwise cosine similarities between gradients of the three objective terms during training. The reported epoch corresponds to the checkpoint used for the gradient comparison on each dataset. Positive cosine similarity indicates cooperative rather than conflicting gradient directions.

Dataset	Grad epoch	L_{pred}-L_{aux}	L_{pred}-L_{focal}	L_{aux}-L_{focal}
ASSIST09	200	0.307	0.498	0.559
ASSIST12	50	0.178	0.433	0.148
ASSIST15	90	0.157	0.588	0.247
NIPS34	130	0.228	0.356	0.389

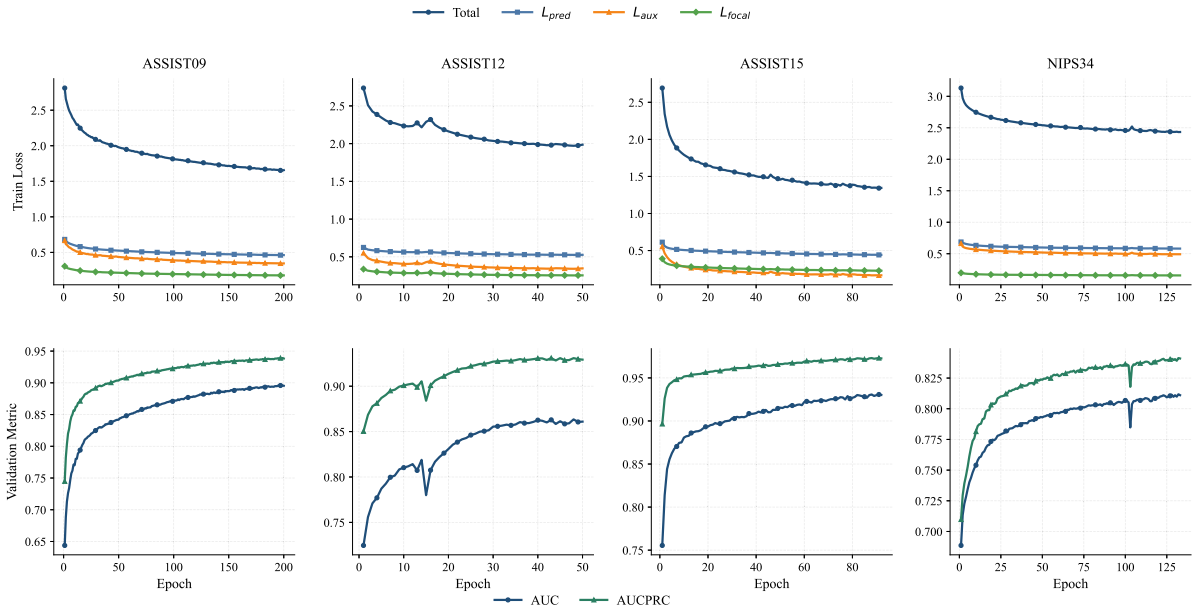


Fig. 5. Objective-stability analysis of CareKT on four benchmark datasets. The top row shows the total training loss together with its three components L_{pred}, L_{aux}, and L_{focal}, and the bottom row shows the validation AUC and AUCPRC trajectories. Overall, the curves suggest stable optimization under the composite objective across all four datasets.

decrease in a stable manner, while the validation metrics improve before stabilizing. Although ASSIST12 and NIPS34 exhibit mild local fluctuations, the overall trajectories remain well behaved and do not suggest optimization instability.

We also summarize pairwise cosine similarities between the gradients of the three loss terms during training in Table 8. The mean cosine similarities remain positive on all four datasets, with (L_{pred}, L_{aux}) ranging from 0.157 to 0.307, (L_{pred}, L_{focal}) ranging from 0.356 to 0.588, and (L_{aux}, L_{focal}) ranging from 0.148 to 0.559. The three objectives therefore appear broadly cooperative rather than strongly conflicting, which is consistent with the stable training behavior in Fig. 5.

6.6. Component ablation and diagnostic robustness (RQ2 & RQ3)

We next examine the contributions of the multi-view structure, auxiliary refinement loss, and focal optimization. We consider one structural ablation, *single-view* ($K = 1$), which keeps all other settings unchanged but uses one local view, and three objective ablations: *w/o FL* removes the focal loss term, *w/o ARL* removes the auxiliary refinement loss, and *w/o ARL+FL* removes both auxiliary loss terms. Table 9 reports the effect of these variants on AUC and ACC, while Table 10 reports AUCPRC for identifying samples from the minority class. Taken together across the four datasets, the complete model provides the most consistent overall performance. CareKT achieves the best AUC on three of the four datasets and the best AUCPRC on three of the four datasets, indicating that the joint use of multi-view decomposition, auxiliary refinement, and focal optimization yields the most consistent balance between predictive discrimination and identification of minority error cases.

The structural ablation shows that replacing the multi-view module with a single local view ($K = 1$) consistently degrades AUC, ACC, and AUCPRC on all four datasets. Relative to this single-view variant, CareKT improves AUC by 0.0354–0.0884 and AUCPRC by 0.0328–0.0538, indicating that the gains are not solely due to the loss design but also depend on preserving multiple local behavioral views.

For the objective ablations, the results suggest that the two auxiliary terms do not contribute in a strictly additive manner. In particular, the relative behavior of the *w/o FL* and *w/o ARL+FL* variants varies across datasets, which indicates dataset-dependent

Table 9
Ablation analysis of the contributions of multi-view decomposition, auxiliary refinement loss, and focal optimization to AUC and ACC. Higher AUC and ACC indicate better performance. Results on all four datasets are reported as averages over the five runs obtained by rotating the validation fold. Here, *single-view* ($K = 1$) is a structural variant that keeps all other settings unchanged but uses one local view. The remaining variants are objective ablations: *w/o FL* removes the focal loss term, *w/o ARL* removes the auxiliary refinement loss, and *w/o ARL+FL* removes both auxiliary loss terms.

Variants	ASSIST09		ASSIST12		ASSIST15		NIPS34	
	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑
Single-view ($K = 1$)	0.7453	0.7186	0.7512	0.7321	0.7523	0.7550	0.7643	0.6986
w/o ARL+FL	0.7504	0.7193	0.8034	0.7545	0.7642	0.7640	0.7727	0.7070
w/o FL	0.7707	0.7364	0.7455	0.7159	0.7271	0.7389	0.7723	0.7078
w/o ARL	0.7881	0.7458	**0.8578**	**0.7893**	0.8050	0.7902	0.7900	0.7220
CareKT	**0.8137**	**0.7641**	0.8396	0.7791	**0.8343**	**0.8027**	**0.7997**	**0.7262**

Table 10
Diagnostic robustness comparison measured by AUCPRC. Higher values indicate better identification of samples from the minority class, namely incorrect responses. Results on all four datasets are reported as averages over the five runs obtained by rotating the validation fold. The ablation variants follow the same definitions as in Table 9.

Variants	ASSIST09	ASSIST12	ASSIST15	NIPS34
single-view ($K = 1$)	0.81618	0.87143	0.87918	0.80508
w/o ARL+FL	0.82203	0.90427	0.89531	0.81397
w/o FL	0.82830	0.86173	0.87790	0.81349
w/o ARL	0.84502	**0.93302**	0.91237	0.82904
CareKT	**0.86996**	0.91772	**0.92864**	**0.83789**

Table 11
Comparison of three fusion mechanisms used in the confidence-aware refinement stage: element-wise multiplication (*mul*), element-wise addition (*add*), and concatenation followed by linear projection (*concat*). Here, *mul* corresponds to the original CareKT design. Reported values on all four datasets are averaged over the five runs obtained by rotating the validation fold. Higher AUC, ACC, and AUCPRC indicate better performance.

Fusion	ASSIST09			ASSIST12			ASSIST15			NIPS34		
	AUC↑	ACC↑	AUCPRC↑	AUC↑	ACC↑	AUCPRC↑	AUC↑	ACC↑	AUCPRC↑	AUC↑	ACC↑	AUCPRC↑
add	0.802	0.758	0.856	0.854	0.791	**0.927**	0.830	0.801	0.927	0.795	**0.728**	0.834
concat	0.806	0.761	0.858	**0.857**	**0.798**	0.927	0.825	0.793	0.924	0.781	0.716	0.825
mul	**0.814**	**0.764**	**0.870**	0.840	0.779	0.918	**0.834**	**0.803**	**0.929**	**0.800**	0.726	**0.838**

interaction between auxiliary refinement and focal reweighting during optimization rather than a simple monotonic effect of removing one term at a time.

Among the two objective components, the focal term shows the clearest contribution to diagnostic robustness: removing it reduces AUCPRC on all four datasets, indicating consistently weaker identification of minority-class error cases. This pattern accords with the original rationale of focal loss, namely down-weighting easy majority examples so that they contribute less to the loss and gradients than harder cases. By contrast, the auxiliary refinement loss is beneficial on most datasets but exhibits less uniform behavior, as shown by the stronger *w/o ARL* result on ASSIST12.

Overall, these findings suggest that focal optimization is the more stable source of minority-error sensitivity, whereas auxiliary refinement provides additional gains in a dataset-dependent manner when combined with the full objective.

6.6.1. Fusion mechanism comparison

To further justify the use of element-wise multiplication in Eq. (12), we conduct an additional fusion-mechanism comparison by replacing the original multiplicative interaction with element-wise addition and concatenation-based fusion. We report averages over the five runs obtained by rotating the validation fold on all four datasets using the same dataset-specific hyperparameter settings.

Table 11 summarizes the resulting AUC, ACC, and AUCPRC comparisons averaged over the five runs obtained by rotating the validation fold on all four datasets. The multiplicative variant performs best on ASSIST09 and ASSIST15. On ASSIST12, the additive and concatenation variants outperform the multiplicative variant, with concatenation yielding the highest AUC and ACC and the additive and concatenation variants showing nearly identical AUCPRC. On NIPS34, the multiplicative variant achieves the highest AUC and AUCPRC, whereas the additive variant achieves the highest ACC. Taken together, these results identify multiplication as the most consistent design across the four benchmarks. This interpretation is consistent with the intended role of confidence-aware refinement: feature-wise multiplication acts as a selective gate on local evidence, whereas simple addition is less selective and concatenation introduces a heavier fusion operator without uniformly better returns.

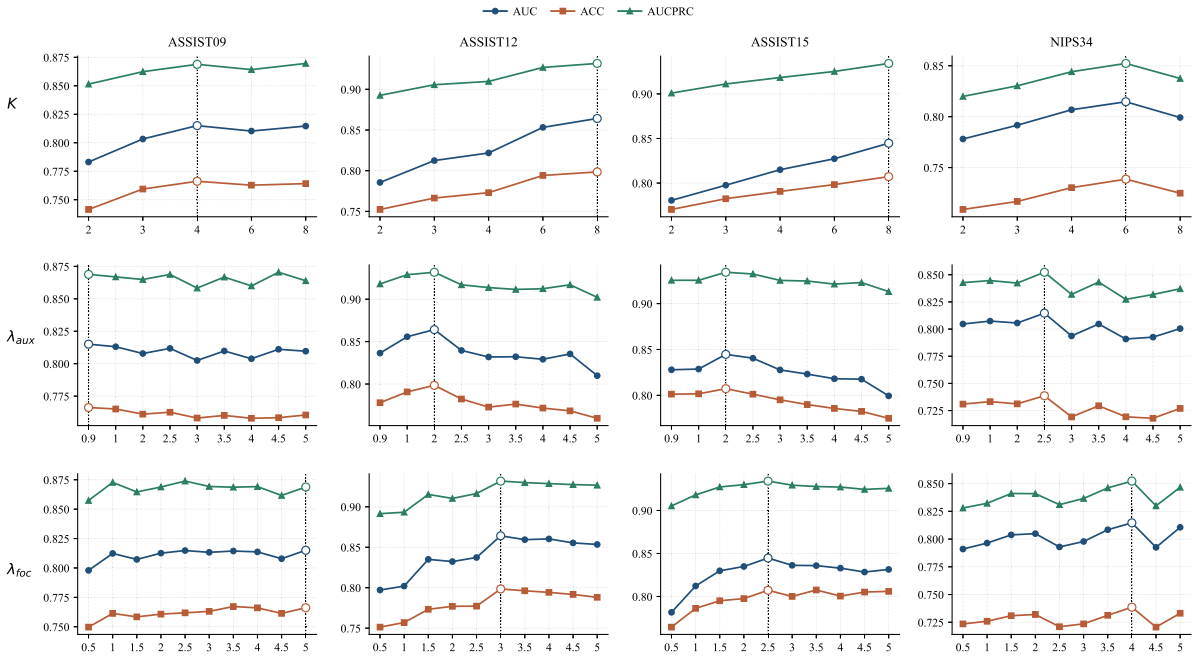


Fig. 6. Sensitivity of CareKT to K, λ_{aux}, and λ_{foc} on four benchmark datasets under the fixed Fold 0 split with seed 42. Columns correspond to datasets, rows correspond to parameter sweeps, and the three curves report AUC, ACC, and AUCPRC. The dotted vertical line and hollow markers indicate the dataset-specific setting used in the main experiments.

6.7. Interpretability and sensitivity analysis (RQ4)

6.7.1. Sensitivity to key hyperparameters

A sensitivity analysis is conducted for three key CareKT hyperparameters by varying one hyperparameter at a time, namely the number of local views K, the auxiliary loss weight λ_{aux}, and the focal loss weight λ_{foc}. To keep this analysis computationally tractable while preserving a consistent comparison across settings, Fig. 6 reports the resulting AUC, ACC, and AUCPRC trends under a fixed sensitivity-analysis protocol using the Fold 0 split and seed 42, while keeping the remaining hyperparameters fixed at the dataset-specific values used in the main experiments. The columns correspond to datasets, the three rows correspond to the sweeps over K, λ_{aux}, and λ_{foc}, and the three curves in each subplot represent AUC, ACC, and AUCPRC, respectively.

The sensitivity curves support three observations. First, moderate increases in K are beneficial, but the gains diminish once the number of views becomes too large. Second, both λ_{aux} and λ_{foc} exhibit broad high-performing ranges rather than sharp isolated optima. Third, the dataset-specific configurations lie at or near the best points on each dataset, namely $(K, \lambda_{aux}, \lambda_{foc}) = (4, 0.9, 5.0)$ for ASSIST09, $(8, 2.0, 3.0)$ for ASSIST12, $(8, 2.0, 2.5)$ for ASSIST15, and $(6, 2.5, 4.0)$ for NIPS34. The performance gains are therefore unlikely to be an artifact of unusually fragile parameter choices.

6.7.2. Diagnostic stratified error analysis on ASSIST12

To better understand the remaining errors on ASSIST12, we conduct a diagnostic stratified error analysis on the CareKT predictions from seed 42 on Fold 0. Specifically, question-level test samples are partitioned in two ways: by the effective history length at the prediction point and by the frequency of the target concept in the training set. In each case, the samples are divided into three quantile-based strata. Accordingly, *Short*, *Medium*, and *Long* denote the low-, middle-, and high-history strata, whereas *Frequent*, *Medium*, and *Rare* denote the high-, middle-, and low-frequency concept strata, respectively. In this analysis, the positive class for false negative rate (FNR) is the incorrect-response class, so a higher FNR indicates more missed minority error cases. The results show a clear dependence on history length. The AUC decreases from 0.8914 in the short-history stratum to 0.7596 in the medium-history stratum and 0.6688 in the long-history stratum, while the corresponding ACC values decrease from 0.8190 to 0.7447 and 0.7097. By contrast, rarity-based stratification yields only a mild AUC variation across frequent, medium, and rare concepts (0.7853, 0.7835, and 0.7760, respectively), although the rare-concept stratum has the lowest ACC (0.7441) and the highest false negative rate (0.0525). These results suggest that the remaining errors on ASSIST12 are associated more strongly with long interaction histories than with concept rarity alone, while low-frequency concepts add further difficulty to the identification of minority error cases.

The corresponding stratified metrics are reported in Table 12.

Table 12
Diagnostic stratified error analysis on the ASSIST12 Fold 0 CareKT predictions. The top block groups samples by history length, and the bottom block groups them by target-concept frequency. Higher AUC, ACC, and AUCPRC are better, whereas lower FNR indicates fewer missed incorrect-response cases, i.e., fewer missed minority error cases.

Analysis	Group	AUC↑	ACC↑	AUCPRC↑	FNR↓
Length	Short	0.8914	0.8190	0.9472	0.0390
Length	Medium	0.7596	0.7447	0.8669	0.0421
Length	Long	0.6688	0.7097	0.8102	0.0414
Rarity	Frequent	0.7853	0.7763	0.8955	0.0288
Rarity	Medium	0.7835	0.7530	0.8792	0.0418
Rarity	Rare	0.7760	0.7441	0.8762	0.0525

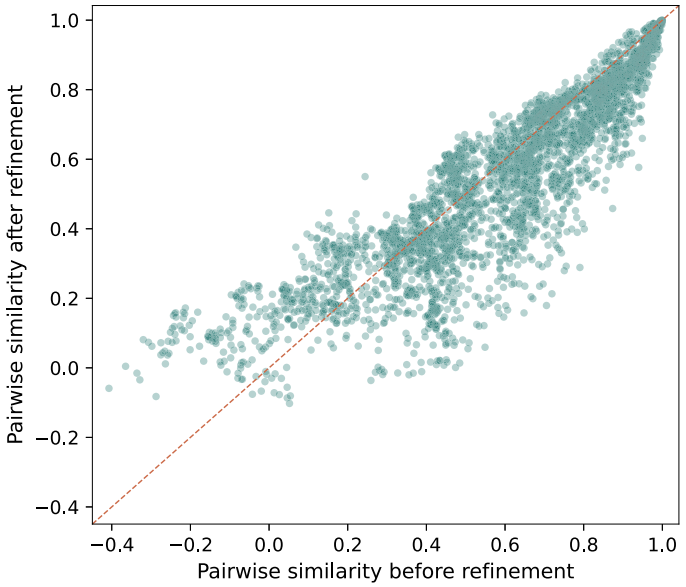


Fig. 7. Pairwise-similarity consistency before and after feature refinement for a sampled learner sequence from ASSIST09. Each point corresponds to one timestep pair. The horizontal and vertical axes show cosine similarity before and after refinement, respectively.

6.7.3. Structural consistency of feature refinement

To assess whether refinement preserves the geometry of the learner representation, we compare pairwise timestep similarities before and after refinement. For a sampled learner sequence from ASSIST09, Fig. 7 plots the cosine similarity of each timestep pair before refinement against the corresponding similarity after refinement. Most points concentrate around the identity line, indicating that refinement changes local features while preserving the dominant pairwise-similarity structure. The quantitative alignment statistics are consistent with this observation. The mean cosine similarity between the original and fused refined representations is 0.705, the correlation between the pairwise similarity structures before and after refinement is 0.888, and the mean absolute similarity shift is 0.098. Together, these results suggest that refinement preserves substantial structural continuity while still producing nontrivial representation updates.

6.7.4. Diversity of multi-view attention patterns

Attention weight distributions produced by different views for a sampled learner sequence from the ASSIST09 dataset are visualized to examine how CareKT uses historical interactions during prediction. In Fig. 8, the horizontal axis indexes the historical interaction positions in the sampled sequence, the vertical axis lists the local-view indices, and the colorbar reports normalized attention weights. Darker green denotes larger attention weights, whereas lighter green denotes smaller weights. The view-specific patterns are nonidentical: different views assign relatively higher attention to different regions of the same interaction sequence. This qualitative example is consistent with the intended role of the multi-view module, although it does not by itself establish view diversity at the dataset level.

To complement this qualitative example, we compute quantitative diversity statistics from the Fold 0 CareKT checkpoint on the test split. Specifically, we measure pairwise Jensen–Shannon (JS) divergence, cosine similarity, top-k overlap with $k = 20\%$ of valid historical positions, and normalized entropy across views. Table 13 shows nontrivial JS divergence (0.353–0.453), low cosine similarity (0.182–0.326), and limited top-k overlap (0.218–0.309) on all four datasets, indicating that the learned views do not

Table 13
Quantitative diversity statistics of multi-view attention patterns on the test split. Higher JS divergence and lower cosine similarity/top-k overlap indicate stronger differentiation across views. Normalized entropy is reported as a complementary descriptor of attention spread.

Dataset	JS Div.↑	Cos Sim.↓	Top-k Overlap↓	Entropy
ASSIST09	0.353	0.270	0.236	0.696
ASSIST12	0.450	0.217	0.273	0.551
ASSIST15	0.378	0.326	0.309	0.570
NIPS34	0.453	0.182	0.218	0.689

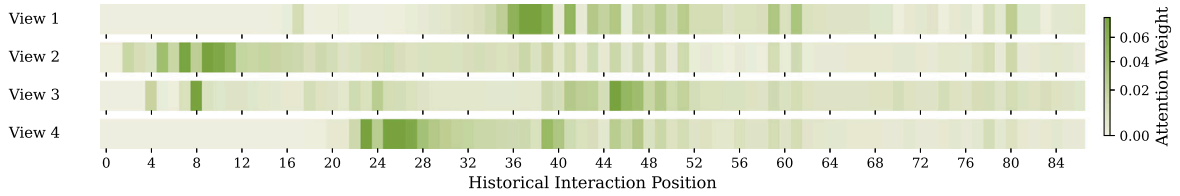


Fig. 8. Attention weight distributions across multiple views in CareKT for a sampled learner sequence from ASSIST09. The horizontal axis denotes historical interaction positions, the vertical axis denotes view indices, and the colorbar denotes normalized attention weights, with darker green indicating larger values.

collapse to identical attention patterns. The normalized entropy remains in a moderate range (0.551–0.696) and is reported as a complementary descriptor of attention spread rather than a direct measure of across-view differentiation.

Fig. 8 provides a qualitative illustration, and Table 13 provides dataset-level quantitative support for the view-specific differentiation learned by the multi-view module over the same interaction history.

7. Discussion

Overall discrimination does not fully characterize diagnostic value in knowledge tracing. AUC remains informative about ranking quality, but it is less revealing of whether a model can reliably identify the less frequent learner errors that matter most for intervention. The AUCPRC evidence is therefore particularly informative, as it suggests improved sensitivity to minority error cases on three of the four benchmarks even though ASSIST12 shows a different tradeoff between the auxiliary refinement term and minority-class retrieval.

The ablation results clarify the complementary roles of the structural and objective components. The multi-view module preserves differentiated local behavioral evidence, the refinement mechanism modulates local features according to view-specific predictive profiles before aggregation, and the auxiliary global-local predictor provides additional supervision during training. The focal loss term acts on optimization by limiting the dominance of the majority class and increasing the contribution of difficult and infrequent examples. The ASSIST12 error analysis also suggests that the gains may vary with dataset characteristics, including sequence density and response distribution. The diagnostic stratified analysis in Table 12 is consistent with this interpretation and suggests that the remaining errors on ASSIST12 are associated more strongly with long interaction histories than with concept rarity alone. Taken together, the present results highlight the importance of sequence modeling, reliable local representations, and training objectives that remain sensitive to diagnostically important learner errors.

7.1. Limitations

Several limitations should be noted. First, some key hyperparameters of CareKT, including the number of local views K and the loss-related coefficients, are selected separately for each dataset according to validation performance. Although the sensitivity analysis places the adopted settings in relatively stable regions, the current configuration remains dataset dependent rather than fully adaptive. Second, the evaluation is conducted on four public KT benchmarks and mainly focuses on concept-level interaction sequences in math- and skill-oriented settings. The generalization of CareKT to broader educational scenarios, such as other subject domains, different interaction structures, or richer input modalities, remains to be established. Third, although CareKT performs consistently on the four benchmark datasets considered here, the current evaluation does not cover settings with substantially more severe class imbalance (e.g., imbalance ratios greater than 5). The robustness of the proposed framework under more extreme minority-error scarcity therefore remains an open question. Although a diagnostic stratified analysis is provided for ASSIST12 in the current evaluation, broader validation and additional fine-grained analyses would further clarify the scope and applicability of the framework.

8. Conclusion

This paper presents CareKT, a hierarchical framework for improving the diagnostic reliability of knowledge tracing under heterogeneous learner behaviors and imbalanced response distributions. CareKT combines global sequence encoding with multiple local behavioral views instead of compressing the interaction history into a single learner representation. The probabilities estimated by individual views provide both the averaged primary prediction and the refinement signals for local representations. An auxiliary predictor that integrates global and local evidence provides additional training supervision, while focal optimization increases the contribution of difficult minority error cases. Experiments on four public knowledge tracing benchmarks show that CareKT achieves the best AUC on ASSIST09, ASSIST12, and ASSIST15 and remains competitive on NIPS34. AUCPRC results, together with precision, recall, and F1-score analyses for incorrect response identification, further indicate that CareKT improves the identification of diagnostically important learner errors on most benchmarks. Ablation, significance, efficiency, sensitivity, and attention diversity analyses indicate that the gains do not come from a single metric or component alone, but from the joint effect of preserving multiple behavioral views, refining local evidence according to predictive confidence, and accounting for class imbalance during training. The NIPS34 results and the ASSIST12 stratified error analysis also indicate that performance remains affected by dataset characteristics, including sequence length and available item information.

Future work will explore extensions that incorporate richer exercise content, structured prerequisite relations, and broader learning environments, and examine whether external educational priors, including those derived from large language models, can further support reliable diagnosis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compeleceng.2026.111411.

Data availability

The datasets used in this study are publicly available benchmark datasets in the knowledge tracing community, including ASSIST09, ASSIST12, ASSIST15, and NIPS34. The anonymized repository for CareKT is publicly available at https://anonymous.4open.science/r/CareKT and contains the code used in this study.

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